BCSE – 3rd year – 1st Semester – 2026-27
Assignment – III [CO4: Q1-Q3, CO3: Q4-Q5]
Operating Systems Laboratory

1. Considering the following scenario, accommodate the processes P4 (350 bytes), P5 (500 bytes) and P6 (150 bytes) respectively (arriving consecutively) using (i) first fit, (ii) best fit strategy, and (iii) worst fit strategy.

Name	Size (bytes)	Starting memory address
P0	800	100
P1	300	1210
P2	550	1870
P3	700	2520

Assume that each memory location is capable of holding only 1 byte and the first location in memory is 0001 and the last address in memory is 9999.

Your output should show the following:

a) The entire memory map including the location and size of each process and hole: i. After P0 to P3 are accommodated, ii. After P4 is accommodated, iii. After P5 is accommodated, and iv. After P6 is accommodated.

b) The entire memory map including the location and size of each process and hole: i. After P0 to P3 are accommodated, ii. After P6 is accommodated, iii. After P5 is accommodated, and iv. After P4 is accommodated.

c) The entire memory map including the location and size of each process and hole: i. After P0 to P3 are accommodated, ii. After P5 is accommodated, iii. After P4 is accommodated, and iv. After P6 is accommodated.

d) The entire memory map including the location and size of each process and hole: i. After P0 to P3 are accommodated, ii. After P6 is accommodated, iii. After P4 is accommodated, and iv. After P5 is accommodated.

2. Write a program to simulate Least Recently Used (LRU) page replacement algorithm for the following page reference string: 9, 10, 11, 7, 12, 8, 7, 6, 12, 5, 4, 3, 10, 11, 12, 4, 5, 6, 9, 4, 5. Consider (i) 5 frames and (ii) 6 frames. Compare the results. Implement using two different methods for finding out the Least Recently Used victim page.

3. Using the above page reference string [refer to Q2] write a program to simulate First In First Out (FIFO) page replacement for a memory with (i) 4 frames, and (ii) 7 frames.

4. Using the above page reference string [refer to Q2] write a program to simulate Least Frequently Used (LFU) page replacement for a memory with 4 frames. Implement using two different methods for finding out the Least Frequently Used victim page.

5. Write a program to simulate *Indexed File Allocation* technique.

6. Write a program to simulate *Linked File Allocation* technique.

Date for submission and viva latest by: November 16th / 18th /19th 2026 (all groups)